

Adding a reusable feature channel beside a prediction service you
cannot change:
an evaluated release problem, one confirmed effect, and three
negative method results

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Abstract

An organisation already publishes a prediction service—fixed probability vectors that downstream recipients consume and cannot be asked to re-accept. We formalise and evaluate what happens when it adds a *second*, reusable feature channel for one of those recipients without altering a single published number, and tries to limit what that recipient, and that recipient colluding with another, can *newly* infer about sex and race. The contribution is an implemented release interface with a measurement contract—per-recipient views, a coalition view, independently fitted attackers, and incremental disclosure always reported beside absolute disclosure—together with what three completed studies of one mechanism family measured through it. Four findings. (i) Exact preservation of the published outputs is structural and holds bitwise on all 210 released views; it does not imply stable service *accuracy*, which moves by up to 0.015 nats across one year. (ii) On a locked, previously unused survey year, a penalty built from the coalition’s joint view beats both an equal-strength and an equal-total-mass local control under simultaneous 95% intervals: 14 of 16 sensitive cells strictly better, none worse, two unresolved. The paired residence constraint is a *point* rule, and interval non-inferiority at the same margin holds in 0 of 4 comparisons. (iii) Two successive refinements of the mechanism—a penalty nonlinear in the channel, then an exactly rotation-invariant repair of that penalty—both fail. The repair is verified exact to 10^{-15} , and the release it produces discloses significantly *more* than the defective objective it repairs on 8 of 48 sensitive cells, with none significantly less. (iv) The first executed external-method comparisons put adapted LEACE, SPLINCE and two OptNet-ARL policies below the developed mechanism on every sensitive endpoint they resolve—31 of $32 \text{ arm} \times \text{endpoint} \times \text{weighting}$ cells; adapted LEACE also costs 0.0098 nats of authorised residence capability against the channel it transformed, a significant loss. We therefore report an empirical and problem-formulation contribution and three negative method results, and we list every claim of our own that this revision withdraws or narrows.

1 What the paper does

Suppose a company already publishes two prediction services: a credit-style model that returns probabilities to one business partner, and a coverage model that returns probabilities to another. Both partners have built systems on those exact numbers, so the company cannot revise them. It now wants to give the first partner something more useful—a compact reusable feature vector built from what the published predictions leave unexplained—while making sure that the partner, and the two partners pooling what they hold, cannot use it to work out an applicant’s sex or race any better than the published predictions already allow. That is the problem this paper implements and measures.

Contribution, precisely. We build the interface and the measurement contract, and we report what a bounded programme of experiments measured through them.

- **The interface.** A mechanism appends a channel Z to one recipient’s published output and modifies

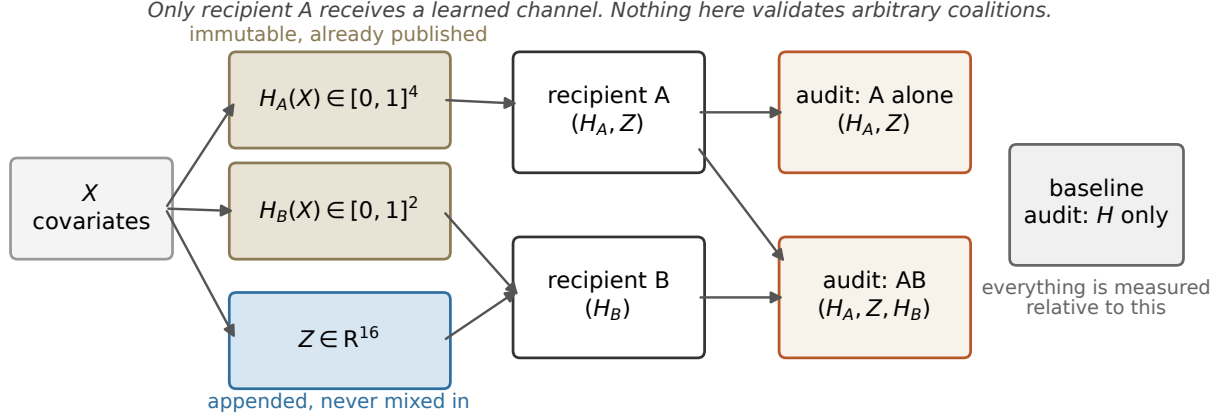


Figure 1: The release graph. H_A and H_B are already published and are never modified; Z is appended to A’s wire alone. Three audit views are scored: A alone, the coalition AB, and the H -only baseline that every number in this paper is measured against. This is a schematic of the interface, not a result.

nothing. Preservation is then an *identity*, not a bounded perturbation, and is checked bitwise on every released view.

- **The comparisons it enables.** Because Z is separable, heterogeneous mechanisms become directly comparable at matched width on the same wire: spectral moment penalties, an adversarially trained neural channel, and closed-form linear erasers, all evaluated by the same independently fitted attacker slate against the same recipient-specific and coalition-specific audit roles, with H -only recovery always subtracted and always also shown unsubtracted.
- **The empirical finding.** On a sealed year, a penalty built from the coalition’s joint view lowers measured recovery relative to two local-only controls matched on penalty strength and on total penalty mass. Everything the project then developed to close the remaining gap to the strongest comparator failed, and three adapted published methods beat it.

This is an empirical and problem-formulation paper, not a method paper. We do not present a deployable mechanism. We are not offering new mathematics: known concept erasure, eigenvalue-sign rank selection, minimax adversarial training and conditional-moment characterisations are prior work, cited as such. Adversarially protected representations that transfer to new tasks are not new either [Madras et al., 2018]. What we did not find, after a bounded search, is a prior *evaluation* of recipient-specific and combined-access policies over an already-published output that the mechanism may not touch (§4); that is absence of evidence after a bounded search, not a priority claim.

2 The interface, and four distinctions we hold to throughout

Each person has covariates X , sensitive attributes S (sex, 2 classes; race, 9 classes) and task labels. Two services are frozen and published before anything here happens: $H_A(X) \in [0, 1]^4$, the full income-over-\$50k and employment probability vectors released to recipient A, and $H_B(X) \in [0, 1]^2$, the coverage vector released to recipient B. A mechanism may add $Z \in \mathbb{R}^{16}$ for A only. So A sees (H_A, Z) , B sees H_B , and the coalition AB sees (H_A, Z, H_B) . H without a subscript means the no-channel interface. The policy authorises A for income, employment and residence and B for coverage and commute, and forbids sex and race everywhere plus nine cross-role combinations—eleven forbidden roles in all. The adversary holds labelled data from the release year, fits fixed attack families on its own pool, selects on a validation pool, and sees only its legal view.

Three names recur. J is a 16-dimensional adversarially trained channel recovered from earlier work and never retrained; it is the strongest comparator in the study and it is *internal*. $L1$ and $L2$ are local controls whose penalty uses only A’s own view, matched to the candidate on penalty strength and on total penalty

mass respectively, so that a coalition effect cannot be confused with “more penalty”. $C1$ adds a penalty term built from the coalition’s joint view; it is the candidate.

Four distinctions carry the argument and are never merged.

Preservation is architectural; accuracy is empirical. Z is concatenated to H_A and never mixed into it, so H is preserved by construction—a property of concatenation, not a theorem. It does *not* follow that the service stays accurate: on identical released vectors, employment log loss rises by 0.0138 nats and income falls by 0.0145 between the two years (Appendix C).

Added capability is measured against the correct H -only view. A channel “adds capability” only if independently fitted probes do better with (H_A, Z) than with H_A .

Added disclosure is a finite-auditor log-loss gain, not an information bound. Every recovery number is the improvement a validation-selected attacker achieves over the same attacker family restricted to H . It is a floor on what is recoverable and never a ceiling (§9).

Recipients have different permitted tasks. AB’s authorised union is not either singleton’s permission set, and the audit roles differ accordingly.

Two scope limits bound every claim below. Only recipient A receives a channel: nothing here demonstrates joint optimisation of channels for several recipients, and the coalition penalty is a constraint on one channel given another recipient’s fixed view, not a multi-channel design. And although residence and commute labels are excluded from every representation fit—the sense in which residence is held out—residence outcomes have been inspected repeatedly across this line of work, so residence is not a pristine unseen task and we do not claim unseen-task generality.

3 The mechanism family, and what is a theorem here

Let T be a frozen PCA32 teacher. Build $\phi(T)$ from standardised T and 96 random Fourier features [Rahimi and Recht, 2007]; residualise against a degree-2 polynomial basis in the released service probabilities and eigen-whiten to V with $V^\top V/n = I$. Utility is $U = C_{VR}C_{RV}$ for the residualised teacher R : for $W^\top W = I$, $\text{tr}(W^\top U W)$ is exactly the training reduction in least-squares reconstruction error of R . Protection matrices P_j are built from household-grouped 3-fold cross-fitted nuisances $m_j(H)$ [Chernozhukov et al., 2016] and the residuals $\text{onehot}(S_j) - m_j(H)$, aggregated over local roles, coalition roles and a marginal variant. The released map is the top-16 eigenvectors of $U_{\text{norm}} - \lambda P$. Inference needs only T and H_A ; H_B enters fitting and auditing and is never an inference input.

Study 2 replaced the fixed penalty matrix by one computed from degree-two monomials of the channel output, so the objective is no longer $\text{tr}(W^\top A W)$ for any W -independent A , and added a predeclared rank-8 variant. Study 3 replaced that penalty again with two blocks that are exactly invariant under $W \mapsto WQ$ for orthogonal Q : a Frobenius-weighted quadratic moment block, for which rotation acts as $M \mapsto Q^\top M Q$, and an exact average of three Gaussian kernels on a frozen 512-row subset, which sees the data only through pairwise distances.

What is proved, and over what. Four statements are elementary and we prove them in Appendix A; nothing else in this paper is a theorem.

1. Appending Z without modifying H preserves H exactly. This is a property of concatenation.
2. An unrestricted recipient holding $[H, Z]$ can ignore Z , so its optimal population log loss is no worse than with H alone. This bounds the *population optimum*, not any fitted model, and our measured increments are sometimes negative.
3. Under unrestricted Bayes prediction and ordinary log loss, the population risk difference between H and (H, Z) equals $I(S; Z | H)$. Our validation-selected finite attackers do not attain those infima, so the reported differences bound that quantity in neither direction, and mutual-information estimation has its own formal limits [McAllester and Stratos, 2020].

4. For the repaired quadratic block, orthogonal reparameterisation preserves the Frobenius norm, and an exact radial kernel is invariant to the corresponding distance-preserving rotations. Both hold identically, subject to stated finite-precision conditions; measured, the objective moves by at most 5.6×10^{-17} under random rotations.

Ky Fan’s maximum principle [Fan, 1949] gives a restricted global optimum for the *original* fixed-matrix family only. The refinements have no fixed matrix to take a spectrum of, so they inherit no optimality statement whatever, and a suboptimal solution remains a live explanation for their results. Eigenvalue-sign rank selection is SARL’s Theorem 3, OptNet-ARL’s Theorem 4.1 and K-TOpt’s Corollary 4.1 [Sadeghi et al., 2019, 2021, 2022]; we fixed $r = 16$ to match the width of the neural comparator, which is a comparison choice, not an optimum. The moment condition is a finite fitted subset of the L^2 characterisation of conditional independence used by kernel CI tests [Zhang et al., 2011, Strobl et al., 2019]—Lemma 2(v) of Zhang et al. [2011], attributed there to Daudin—quantified over $f \in \{Z_\ell b_k(H)\}$ rather than over every square-integrable f . Vanishing fitted moments therefore establish no conditional independence, and a non-zero moment may reflect nuisance error: one of our own fixtures shows a misspecified $m(H)$ manufacturing a penalty more than ten times the oracle value under *exact* conditional independence.

4 Relation to prior work

The closed-form line of Sadeghi, Dehdashtian and Boddeti already solves this shape of problem. SARL [Sadeghi et al., 2019] takes eigenvectors of a utility-minus- λ -dependence matrix; OptNet-ARL [Sadeghi et al., 2021] keeps closed-form ridge players inside a learned encoder and states in one sentence that it generalises to several sensitive attributes; K-TOpt [Sadeghi et al., 2022] adds random Fourier features and evaluates on Folktables [Ding et al., 2021]; U-FaTE [Dehdashtian et al., 2024] adds a *conditional* penalty. Our marginal arms are, up to parametrisation, SARL’s construction on whitened residuals—verified by bitwise rebuild (§8). Adversarial fair *and transferable* representation learning is Madras et al. [Madras et al., 2018], and we claim nothing new for that combination. Linear concept erasure [Belrose et al., 2023, Holstege et al., 2025] supplies an exact guarantee against *linear* adversaries on a representation the releaser owns and may edit. Adversarial removal is known to leave recoverable information [Elazar and Goldberg, 2018], and least-privilege learning has fundamental limits [Stadler et al., 2024]. **Reusing a known solver, a conditional-moment characterisation or a rank rule is attribution, not innovation.**

Three things differ, with their weight stated honestly. *What we condition on:* U-FaTE’s conditional measure is stratification over a discrete label, and H_A is a pair of continuous probability vectors, so we substitute cross-fitted nuisance residuals; H is also a *released artifact* rather than the label to be predicted. *What utility means:* reconstruction of a residualised teacher rather than dependence with a label. *Recipient structure:* separate local and coalition penalties over a fixed prior output, with only one recipient receiving a channel. The first two are adaptations of known components. For the third, the nearest candidate we found is weaker prior art than its title suggests: the multi-consumer information-theoretic line [Sankar et al., 2013] designs one sanitised release consumed by many users, and we could not confirm from its text that it carries a per-recipient release or a collusion term—we mark that entry ambiguous rather than assert it. Appendix B gives the full comparison table over five features (fixed pre-existing output, reusable extra features, recipient-specific permissions, coalition-aware training, independent attack evaluation), with every entry we could not verify from a full text marked as unverified rather than guessed.

5 Evidence status, and a chronology that keeps four studies apart

California ACS one-year PUMS, ages 19–34, three household splits throughout, with disjoint pools for representation fitting, source validation, task fitting and validation, attacker fitting and validation, and evaluation. Attacks are logistic regression, two restarted MLPs at 120/360 epochs, two gradient-boosted trees and a random-feature kernel-ridge family, plus every legal projection of H ’s own attacks and of singleton candidates into the coalition view. Uncertainty is a paired household-cluster bootstrap with single-step max- $|t|$ simultaneous intervals inside each pre-declared family.

Study	Commit	Data pools	Evidence role	Registration	Outcome
1. Locked 2017 transport	349efa4	2018 fit → 2017 sealed	Confirmatory	prespecified rules, no directional forecast	coalition penalty beats both local controls
2. Nonlinear / rank refinement	c37807e	2018 development; 2017 reuse	Development	6 forecasts, 3 wrong or ambiguous	moved recovery, failed 8/8 coordination; objective later found coordinate-dependent
3. Invariant repair + externals	73903b7	2018 development; 2017 reuse	Development	7 forecasts, 3 wrong	repair exact; measured disclosure worse; three external arms beat it
4. Direct adversarial refinement	<i>pending</i>	<i>to be read from the committed matrix</i>	<i>Development</i>	<i>pending</i>	<i>pending</i>

Table 1: Four studies, kept apart by what each is evidence *for*. Only Study 1 is confirmatory. Study 3 is a *new formulation designed to repair a defect Study 2 diagnosed after evaluation*; it does not retroactively repair Study 2’s evaluated maps, and its 2017 pass is development on a partition Study 1 already spent, not a second sealed-year confirmation. ACS 2016 is admitted on provenance and schema only and remains **unscored**: no label, output, transform or performance figure for that year has been inspected by any study.

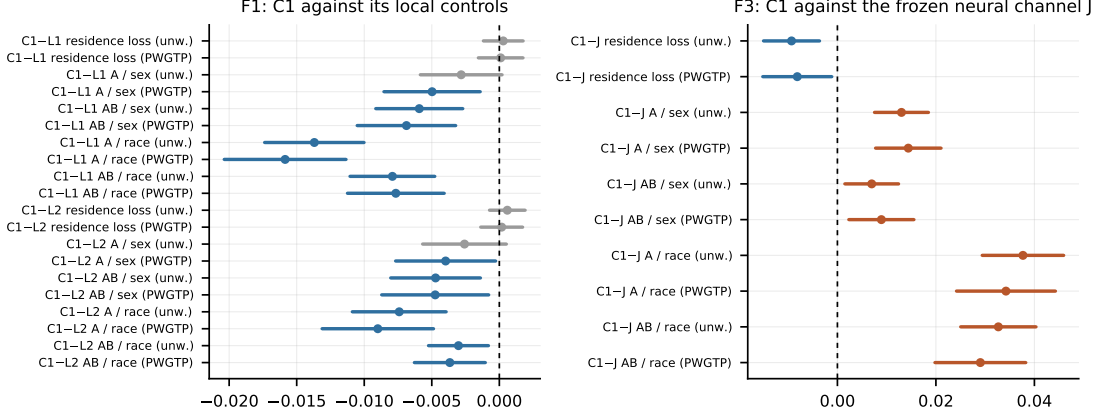
The distinction Table 1 enforces is worth stating in words, because it is easy to blur. Study 2 fitted and evaluated maps, and *afterwards* diagnosed that its penalty was coordinate-dependent. Study 3 wrote a new objective that is provably free of that defect, and refit under it. Those are three different things: a diagnosed bug, a new formulation, and a rerun. The Study 2 numbers in this paper remain numbers for the defective objective and are labelled so; the Study 3 numbers are numbers for a different construction, not corrected versions of Study 2’s.

6 Study 1: the confirmed coalition effect, and the rule it did not establish

On the sealed 2017 partition, *C1* has a simultaneous-interval advantage over both *L1* and *L2* under both weightings, and the registered **advantage** rule fired in all four contrast $\times$ weighting cells. The effect is real and not uniform: of the 16 sensitive cells in family F1, **14 have adjusted intervals entirely below zero, none is above zero, and two are unresolved** (*C1* versus *L1* on A/sex unweighted, $[-0.00585, +0.00020]$, and versus *L2*, $[-0.00568, +0.00052]$). Frozen 2018 attacks reproduce the ordering, so it is not an artifact of refitting attackers on the new year. Every augmented interface clears the 0.01-nat residence reference over *H* in every seed, weighting and mode.

The paired residence rule is a point criterion and only a point criterion. As implemented it is $\text{seed-mean}(C1 - \text{comparator}) \leq 0.001$ on residence log loss; the four values are 0.00029, 0.00010, 0.00059 and 0.00017, so it passes. It does *not* follow that the cost is below 0.001: **zero of four comparisons support interval non-inferiority at that margin**, missing it by 2% to 35% even under the most permissive procedure we computed, and zero support two-sided equivalence (Appendix C). “No statistically detected cost” and “a cost bounded below 0.001” are different statements and only the first is supported. The same applies to the harm side: the **advantage** rule requires that no sensitive endpoint be significantly worse, which is an absence-of-evidence condition, and on the two unresolved A/sex cells up to 5×10^{-4} nats of *extra* local sex recovery is not excluded.

And the mechanism is not competitive with *J*. *C1* is significantly worse than *J* on all four sensitive endpoints under both weightings and on both source probes, and better only on residence, $-0.0093 [-0.0150, -0.0036]$. That ordering survives matching the two channels on average utility in both directions (Appendix C). It does *not* make *J* dominant: *J* carries materially less residence capability than every spectral arm, so neither channel Pareto-dominates the other, and *J* is an internal channel, so this is evidence about *J* and not about the state of the art.



paired difference (nats); left of zero favours the candidate. Blue: interval entirely below zero. Orange: entirely above. Grey: crosses zero (unresolved, not unaffected).

Figure 2: Paired differences with simultaneous 95% intervals inside each pre-declared family, fresh 2017 attacks, both weightings, candidate minus comparator in nats. **Left, F1:** *C1* against its two local controls. The two grey *A/sex* rows are *unresolved*—the comparison is undecided, not the endpoint unaffected. **Right, F3:** *C1* against *J*.

7 Studies 2 and 3: two refinements, both negative

Study 2 asked whether that gap closes if the penalty acts on nonlinear functions of the channel, or the released rank is reduced, or both. Both factors moved measured recovery in the intended direction and they act on different attributes—the nonlinear penalty most on sex, rank-8 compression most on race—taking additional *A/race* recovery from 0.0274 to 0.0055 against *J*’s 0.0071 while holding residence gain at 0.0259 against *J*’s 0.0215. But **all eight registered coordination cells failed**, the best candidate *n1r8-C1* remained significantly worse than *J* on local sex (+0.01065 unweighted, +0.01210 person-weighted; adjusted intervals exclude zero), an exploratory cross-year look widened rather than narrowed that gap, and the nonlinear penalty was not harmless: at rank 8 with the *L1* policy it made *A/race* recovery significantly *worse*, +0.00862 [+0.00168, +0.01556] unweighted. Denominators are in Appendix D; most of the grid is simply undecided at three seeds.

Study 2 then diagnosed, after those maps had been evaluated, that its penalty was *coordinate-dependent*: summing degree-two monomials $a \leq b$ with equal weight computes $\sum_a M_{aa}^2 + \sum_{a < b} M_{ab}^2$ rather than $\|M\|_F^2$, and only the latter survives $M \mapsto Q^\top M Q$. A separately optimised, information-preserving rotation of the original solution’s subspace reaches a mean 0.628 of the refinement’s training-objective gain (range 0.367–0.907 over 18 conditions). That share is **attained, not a supremum**—every rotation search stopped on budget exhaustion or a failed line search—and it is a property of the objective, not an accounting of what the optimiser did.

Study 3 repaired it exactly, and the repair made the measured outcome worse. The repaired objective is a function on the Grassmannian: rotation-only share 1.4×10^{-15} over 18 cells against a registered forecast of 0.01, and $|L(WQ) - L(W)| \leq 5.6 \times 10^{-17}$ on real data. This gate was computed after fitting and *before* any 2018 or 2017 score was read, which is why the repair claim is supportable at all. Under the repaired objective—the only common yardstick—the repaired map attains a lower value than the defective one in 18 of 18 matched cells. And its releases disclose *more*: over six matched contrasts $\times$ four sensitive endpoints $\times$ two weightings, **8 of 48 cells are significantly worse and none is significantly better** (Table 2). The largest is +0.0107 [+0.0034, +0.0180] on local race at rank 16; none of the twelve residence cells resolves in either direction, so the extra disclosure buys no measured capability. Of Study 3’s seven registered forecasts, three were wrong, and the authors’ own registration of the central one was ill-posed; Appendix E reproduces all seven.

What this does and does not show. It refutes the prediction made from the rotation diagnostic—that removing the provably inert slack would let the same compute buy more protection. It does *not* show that rotation invariance is harmful, and it does not isolate a cause. The repair changed five things at once:

Quantity	Defective objective (Study 2)	Repaired objective (Study 3)
Rotation-only share of training gain, attained (<i>a property of the objective</i>)	mean 0.628, range 0.367–0.907 over 18 cells	max. abs. 1.4×10^{-15} over 18 cells
Direct invariance identity $ L(WQ) - L(W) $ on real data	up to 7.5×10^{-3}	5.6×10^{-17}
<i>Measured disclosure, repaired minus defective, over 6 matched contrasts $\times$ 4 sensitive endpoints $\times$ 2 weightings:</i>		
Sensitive cells significantly worse (more recovery)	8 of 48	
Sensitive cells significantly better (less recovery)	0 of 48	
Largest single increase in recovery	+0.0107 A / race, riv16-L2	
Residence cells resolved in either direction	0 of 12	

Table 2: The repair worked on the objective and did not work on the release. The upper rows are a diagnostic of the *objective*; the lower block is a measurement of the *release*. They are reported separately because an exactly invariant objective did not produce less measured disclosure. The attained rotation share is not a supremum – every rotation search stopped on budget exhaustion or a failed line search – and it is not a causal decomposition of what the optimiser did.

Arm	wt.	additional recovery vs. riv16-C1				residence	source
		A / sex	AB / sex	A / race	AB / race	loss diff.	allow.
LEACE	unw.	−0.0081▼	−0.0086▼	−0.0131▼	−0.0158▼	+0.0036	3/3
	PWGTP	−0.0097▼	−0.0093▼	−0.0154▼	−0.0166▼	+0.0040	
SPLINCE	unw.	−0.0088▼	−0.0077▼	−0.0183▼	−0.0165▼	+0.0089▲	3/3
	PWGTP	−0.0100▼	−0.0084	−0.0201▼	−0.0163▼	+0.0091▲	
OptNet-C1	unw.	−0.0074▼	−0.0110▼	−0.0161▼	−0.0145▼	+0.0086▲	2/3
	PWGTP	−0.0089▼	−0.0117▼	−0.0183▼	−0.0151▼	+0.0076▲	
OptNet-L1	unw.	−0.0032	−0.0048	−0.0056	−0.0060	+0.0057	2/3
	PWGTP	−0.0059	−0.0069	−0.0116	−0.0075	+0.0046	
OptNet-L2	unw.	−0.0104▼	−0.0092▼	−0.0184▼	−0.0198▼	+0.0096▲	2/3
	PWGTP	−0.0125▼	−0.0112▼	−0.0210▼	−0.0212▼	+0.0082▲	

Table 3: Adapted published methods against the repaired mechanism at rank 16 with the coalition policy. Every arm transforms or replaces *only* the auxiliary channel and appends it to unchanged H_A . ▼ adjusted simultaneous interval entirely below zero (the arm recovers *less* than riv16-C1); ▲ entirely above. Residence is a *log loss* difference, so a positive value means the arm supplies *less* residence capability. ACS 2018 development pools; three seeds. The last column counts seeds clearing the legacy service-probe allowance, out of 3 (riv16-C1 clears 1/3).

Frobenius weighting, a single scalar block scale in place of per-feature standardisation, no centering, an exact radial kernel in place of a finite Fourier bank, and frozen per-role 512-row subsets. No ablation separates them, and the kernel substitution changes the estimand—the new block averages over at most 512 rows and carries a measured $1/m$ V-statistic floor. Surrogate–attacker mismatch remains the live explanation; “coordinate dependence was an implicit regulariser” is a hypothesis this design cannot test, and we do not assert it. Nor does four failed coordination cells in one construction show that nonlinear conditional-dependence penalties cannot satisfy a coordination rule. Finally, the repaired arms are significantly *worse* than their own linear-moment controls on A/race at rank 8 in three contrasts, so “the repaired arms beat their own controls”—true of exploratory 2017 seed means—is not true of the 2018 intervals.

8 The first executed external comparisons

Studies 1 and 2 ran no published method. Study 3 did, and the answer is unambiguous in one direction and carefully bounded in the other.

What each adaptation is, and what it is not. LEACE [Belrose et al., 2023] and SPLINCE [Holstege et al., 2025] are applied to the 16-coordinate auxiliary channel *alone* and the transformed channel is appended to unchanged H_A . They are deliberately *not* applied to the concatenated $[H_A, Z]$ wire: LEACE’s P^* is a single oblique projection over all coordinates and nothing constrains it to act as the identity on H_A , so editing

Arm – J	wt.	A / sex	AB / sex	A / race	AB / race	residence
LEACE	unw.	+0.0055	−0.0020	+0.0019	−0.0022	−0.0005
	PWGTP	+0.0056	−0.0016	+0.0034	−0.0027	−0.0010
SPLINCE	unw.	+0.0047	−0.0011	−0.0034	−0.0028	+0.0048
	PWGTP	+0.0052	−0.0007	−0.0013	−0.0025	+0.0041
OptNet-C1	unw.	+0.0061	−0.0044	−0.0012	−0.0009	+0.0045
	PWGTP	+0.0064	−0.0041	+0.0005	−0.0012	+0.0026
OptNet-L1	unw.	+0.0104▲	+0.0017	+0.0093	+0.0077	+0.0017
	PWGTP	+0.0094▲	+0.0008	+0.0073	+0.0064	−0.0004
OptNet-L2	unw.	+0.0031	−0.0026	−0.0035	−0.0062	+0.0056
	PWGTP	+0.0028	−0.0035	−0.0022	−0.0073	+0.0032
riv16-C1	unw.	+0.0135▲	+0.0066	+0.0149▲	+0.0136▲	−0.0041
	PWGTP	+0.0152▲	+0.0077	+0.0188▲	+0.0139▲	−0.0050
riv8-C1	unw.	+0.0121▲	+0.0058	+0.0063	+0.0049	−0.0049
	PWGTP	+0.0147▲	+0.0070▲	+0.0099	+0.0078	−0.0052

Table 4: The same arms against the frozen neural channel J . A blank cell is an **unresolved** difference at three seeds, not a demonstrated equality: the adjusted interval contains zero and also contains effects of the size this table reports elsewhere. No equivalence or non-inferiority test was run for these cells. Negative residence means the arm supplies *more* residence capability than J .

the joint wire would generally change the published service output. The consequence for interpretation is the important part. **Their empirical linear guarantee applies to the fitted erasure of that channel, on the moments it was fitted on. It is not a guarantee about arbitrary nonlinear prediction from the concatenated service-plus-channel view an attacker actually holds**—LEACE’s own authors conjecture that nondestructive editing against a general nonlinear adversary is intractable, and a fixture in this study confirms the boundary concretely: after erasure a linear probe reads $R^2 < 10^{-8}$ while a quadratic feature of the same erased channel still recovers. SPLINCE additionally preserves covariance with the *authorised training* tasks, not with the held-out residence task; that is an adaptation, and an asymmetry in SPLINCE’s favour on utility, since no other arm sees a task label during representation fitting. SPLINCE also shares LEACE’s kernel, so under its own Theorem 2 the two coincide after refitting an unregularised unique-minimiser model—any difference measured here comes from attacker regularisation and nonlinearity, never from a stronger guarantee. OptNet-ARL’s multi- λ multi-attribute form is an **extrapolation**: the paper asserts multi-attribute generalisation in one sentence with no equation, no normalisation rule and no code. And the project’s marginal spectral arms turned out to *be* SARL-style rebuilds—reconstructing them from the derivation reproduces the stored maps bitwise—with one measured mismatch that is not absorbable into $\bar{\lambda}$: per-attribute trace reweighting, moving the selected subspace by projector Frobenius distance 0.10–0.20. They are credited and reused, and six duplicate fits were avoided. **Shared policy coefficients do not imply equal effective regularisation across methods.**

The result. Adapted LEACE, OptNet-C1 and OptNet-L2 recover significantly less than the repaired spectral mechanism on all four sensitive endpoints under *both* weightings, and adapted SPLINCE on seven of those eight cells (Table 3; its coalition-sex cell is unresolved under person weighting). OptNet-L1 separates on none. **A closed-form linear eraser from 2023, applied to a channel this project already had, recovers less than the mechanism this line of work developed across three studies.** That is the consequential finding, and it is a finding about our mechanism.

The utility side must be stated with equal care, and the incoming study’s own summary overstated it. LEACE’s residence-loss difference against the repaired arm is +0.0036 [−0.0028, +0.0100] unweighted and +0.0040 [−0.0031, +0.0110] person-weighted. That is *unresolved*, not costless: the interval admits a cost of up to 0.010 nats, larger than the recovery advantage LEACE wins on three of the four endpoints, and it contains SPLINCE’s +0.0089, which the same analysis calls significant. Against the channel LEACE actually transformed, the cost is real: +0.0098 [+0.0036, +0.0159] unweighted and +0.0078 [+0.0011, +0.0145] person-weighted, with mean residence gain over H falling from 0.0318 to 0.0220. **Linear erasure of this channel buys its protection with measurable authorised capability.** We therefore do not say LEACE dominates anything.

One panel per sensitive endpoint; no score is averaged across them. Filled markers clear the service-probe allowance in 3/3 seeds, hollow do not.

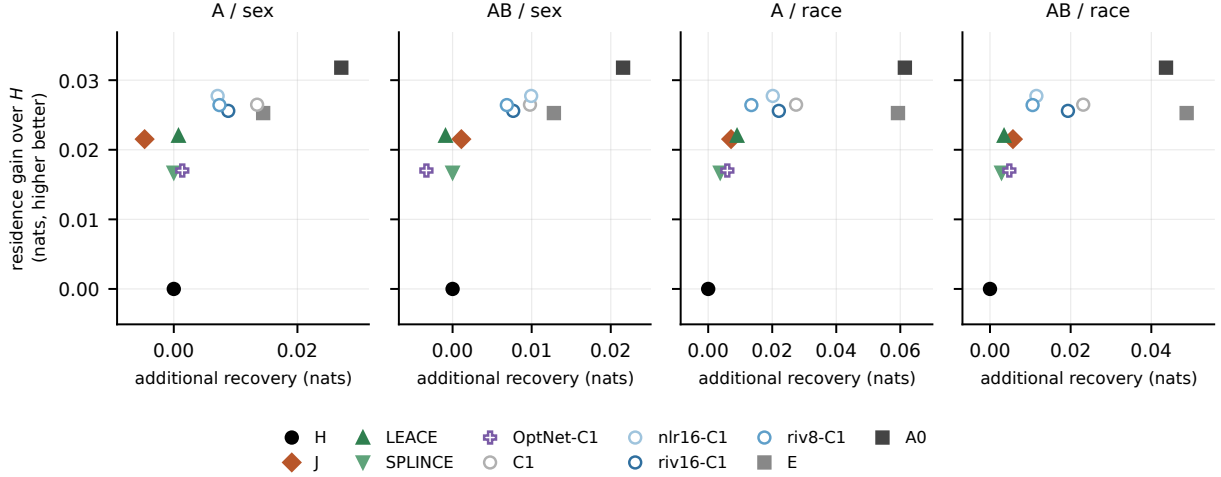


Figure 3: Authorised capability against additional recovery, one panel per sensitive endpoint. No privacy score is averaged across endpoints: doing so would let a favourable sex result hide an unfavourable race result, which is exactly what happens for several arms here.

Against J , LEACE, SPLINCE, OptNet-C1 and OptNet-L2 are *not separated* at three seeds on any of the five family endpoints, while the repaired spectral arm is significantly worse on three of four (Table 4). “Not separated” is not equivalence: no equivalence or non-inferiority test was run for these cells, and all three external point estimates on local sex lean towards J , two of them missing significance by less than 0.0007. OptNet-L1 is significantly worse than J on local sex, so the OptNet family is not uniformly indistinguishable from it either.

One structural finding worth keeping. Linear erasure deletes exactly $\text{rank}(\Sigma_{XZ})$ directions, so the realised width of the released channel is 6, 6, 7 across the three seeds. Seed 2 differs because one race category has *no support in that seed’s own representation-fitting pool* (counts 1, 1, 0); the category is not absent from California, and a class without support is unmeasured, not protected. **Protected-class support in the fitting pool determines the realised width of a released channel** (Appendix E).

9 What this evidence does not establish

Stated once, in one place, rather than scattered.

No attack maximum is an upper bound. Every recovery number is what a *validation-selected* member of a finite attack collection achieved. It is a floor. Our own data contain a clean demonstration: when J is given additional attack candidates, its measured absolute recovery *falls*, because selection on validation loss can pick a candidate that generalises worse (Appendix F). A stronger attacker we did not build would move any of these numbers upward.

Nested validation-selected models need not produce non-negative gains. Statement 2 of §3 is about population optima. In finite samples, 22 of 429 unweighted role-cells have *negative* increments, and J ’s own local-sex increment is negative under person weighting with an interval excluding zero. We report these unclipped, because clipping them would manufacture protection out of a selection artifact. H ’s own attack, routed onto an augmented wire, remains executable, so a negative increment removes nothing an H -only attacker already had.

No detected difference is not equality. Most of the grid in every attribution family is unresolved at three seeds. We mark such cells blank and never read them as parity.

Claim	Status	Supporting measurement	What it does not establish
H preserved exactly in every release	Structural identity	Verified bitwise, 210/210 views	Says nothing about service <i>accuracy</i> , which moved by up to 0.015 nats
Appended channel adds authorised capability	Empirical, development & sealed	0.017–0.032 nats residence gain over H	Half-headroom reference fails in 1–3 of 3 seeds for every arm
Coalition penalty beats both local controls	Confirmatory (2017 sealed)	14/16 cells better, 0 worse, 2 unresolved	One year, one state, three seeds; residence cost not bounded below 0.001
Developed mechanism vs. frozen neural channel	Empirical, development	Worse on 3 of 4 sensitive endpoints	J is internal, and it supplies <i>less</i> residence capability: neither dominates
Rotation defect repaired exactly	Elementary algebra + numerics	Invariance to 10^{-15} , 18/18 cells	Invariance of an objective is not evidence the objective measures the right thing
Repair reduces measured disclosure	Refuted	4 cells significantly worse, 0 better	Attributable to the repaired <i>package</i> , not to any single changed ingredient
External adaptations beat the developed mechanism	Empirical, development	LEACE / SPLINCE / OptNet-C1 lower on all four endpoints	Adaptations, not replicas; LEACE costs residence against its own source channel
Any bound on $I(S; Z H)$	Not claimed	–	Finite validation-selected attacks attain no infimum in either direction

Table 5: Every load-bearing claim, its status, the measurement behind it, and what it does not establish. The full traceable list, with source commit, artifact, endpoint, data pool and uncertainty procedure for each, is `CLAIM_LEDGER.csv`.

Ratios are scale-dependent; paired differences are not. $C1$ against J on A/race is $6.0\times$ on increments and $2.6\times$ in absolute terms; the paired difference is identical on both scales because the shared H baseline cancels inside a paired contrast. We therefore quote differences. This also makes every headline contrast in this paper invariant to the choice of attack scope, even though the H -relative *levels* are not (Appendix F).

Nothing here is a certificate. Vanishing fitted finite moments imply neither $Z \perp S | H$ nor any bound on $I(S; Z | H)$; no Type-I control is inherited from KCI or RCoT; and a misspecified nuisance manufactures a penalty more than ten times the oracle value under exact conditional independence.

Development against confirmation. Only the 2017 transport is confirmatory. The 2018 pools have been used repeatedly; the 2017 seal is spent, so Study 3’s cross-year pass is development, carries no intervals by design, and excludes the erasure and OptNet arms entirely. **ACS 2016 remains unscored:** admitted on provenance and schema with a prospective label-blind split prepared, with no label, output, transform or performance figure inspected. A positive development result may motivate a confirmation; it can never inherit a sealed year’s status.

Sampling and design. Three seeds are three fixed fitted systems on one cohort. The bootstraps cover people, not retraining variability and not Census design variance. Transport is between public survey years of the same state and age band, and cross-year person distinctness cannot be established from public identifiers. Income is nominal and unadjusted and its estimand differs again in 2016.

Numerical integrity. Study 3’s exploratory 2017 stage aborted four times with a probability row summing to exactly zero inside an otherwise well-formed array. Every occurrence was detected by full-schema validation and aborted its unit before any metrics file was written; the four affected units were quarantined and recomputed by the unpatched pipeline, so no corrupted array reached any table. The machine was at swap capacity throughout. **Memory pressure was the observed condition, not a demonstrated cause**, and nothing here claims the machine is now proved reliable. One document in the incoming package records that no such fault occurred; it predates the stage in which they did, and we follow the run ledger instead (`CORRECTIONS.md B10`).

10 Claims this revision withdraws or narrows

Seventeen items were corrected in the previous revision and are carried forward in `CORRECTIONS.md`. Fourteen are new here. The substantive ones:

1. **“LEACE dominates the repaired mechanism with no statistically detectable residence cost.”**
The recovery half holds. The utility half does not: the difference is unresolved with an interval admitting 0.010 nats, it contains SPLINCE’s significant cost, and against the channel LEACE transformed the

cost is significant at +0.0098. *Dominates* is withdrawn.

2. **“LEACE, SPLINCE and OptNet-C1 are statistically indistinguishable from J .”** Narrowed to unresolved differences. No equivalence test was run and all three point estimates lean towards J on local sex.
3. **J is not Pareto-dominant.** It recovers less and supplies less residence capability than several spectral arms, significantly so against `riv16-L1`. It is a trade.
4. **“Coordinate dependence was acting as an implicit regulariser.”** Narrowed to a hypothesis. Five ingredients changed together and no ablation separates them.
5. **The 63% rotation share** is an attained diagnostic value, not a supremum and not a causal decomposition of training benefit.
6. **“Zero population support”** for a race category is a *fitting-pool* count (1, 1, 0 by seed). A category missing from one pool is not absent from the population.
7. **The two carried numerical corrections are kept:** the `n1r8-C1` versus J local-sex deficit is +0.01210 person-weighted, not +0.0098; and “no endpoint significantly worse” for the nonlinear-penalty factor stays withdrawn, the counterexample being +0.00862 [+0.00168, +0.01556] unweighted.
8. **“Terminal 1’s experiment does not exist”** was stale. Study 3 is complete, committed and integrated here at 73903b7f.

11 Conclusion

A service that has already published predictions can add a reusable channel beside them without changing a released number, and can steer that channel away from directions a recipient or a colluding pair would use to infer sex and race. On a locked, previously unused survey year, the coalition-conditioning knob does what it claims against both of its local controls—14 of 16 sensitive cells strictly better, none worse, two unresolved—while costing residence capability by an amount the data cannot bound below the 0.001 nats the registered rule used. Everything built afterwards to make the mechanism competitive failed: a nonlinear penalty failed all eight of its coordination cells and made one endpoint significantly worse; an exact repair of that penalty’s coordinate dependence—the one intervention the diagnostic predicted would help—produced significantly *more* disclosure on eight of forty-eight cells and less on none; and the first executed external comparisons put three adapted published methods below the developed mechanism on every sensitive endpoint.

The defensible contribution is the release problem and the evaluation design: fixed published outputs as a hard constraint, recipient-specific and coalition-specific audit roles, fresh and frozen attackers never pooled, absolute and incremental recovery reported side by side, negative increments retained unclipped, denominators printed beside every count, per-endpoint results never averaged into one score, and a sealed temporal transport with the multiplicity fixed in advance. The measurement contract is what made three successive negative results legible instead of ambiguous. The mechanism family is not competitive, and we say so.

12 Reproducibility

Deterministic seeds throughout; every unit writes a hash-bound completion record. All 90 pre-declared Study 1 endpoints were regenerated from stored per-person predictions by an independently written scorer agreeing to 1.1×10^{-15} . Study 3’s own independent score replay reports a maximum absolute difference of 4.4×10^{-16} over 3,264 checks, and H_A/H_B parity is bitwise on all seven pools for every arm at release time and again at evaluation time. Every table and figure here is generated by two scripts from committed evidence; the Study 3 assets are read at a pinned commit through the object store, so each printed digit traces to a (commit, path, sha256) triple in `MANIFEST.json`. No number was typed by hand. Commands and file map: `REPRODUCE_PAPER.md` in `results/pcrl_manuscript_review_v3/`.

A Elementary results and their boundaries

Proofs and the exact conditions under which each holds are in `MATHEMATICAL_REVIEW.md`. Each statement is given at four levels—population definition, empirical estimator, algorithmic approximation, numerical test—because conflating them is the failure mode this appendix exists to prevent.

B Related-work comparison

Work	fixed prior output	reusable extra features	recipient permissions	coalition-aware training	independent attack eval.
This paper	✓	✓	✓	✓	✓
Madras et al. 2018 (LAFTR)	–	✓	–	–	part.
LEACE [Belrose et al., 2023]	–	–	–	–	–
SPLINCE [Holstege et al., 2025]	–	–	–	–	–
SARL [Sadeghi et al., 2019]	–	–	–	–	part.
OptNet-ARL [Sadeghi et al., 2021]	–	–	–	–	part.
K-TOpt [Sadeghi et al., 2022]	–	–	–	–	part.
U-FaTE [Dehdashtian et al., 2024]	–	–	–	–	✓
Sankar et al. [2013]	–	UNV.	UNV.	UNV.	–
Stadler et al. 2024	–	–	–	–	n/a
Elazar & Goldberg 2018	–	–	–	–	✓

Table 6: The five features this paper’s setting turns on. ✓ the text demonstrates it; – its setting excludes it; “part.” a partial form (an adversarial objective or a transfer test, not an independent attacker slate); UNV. could not be settled from what was read and is **not** guessed from a title or abstract. Several of these methods can be *adapted* into this setting; three were, and they beat the mechanism developed here. Provenance for every row, including which sources were read in full text and which only as abstracts, is in `RELATED_WORK_COMPARISON.md`.

The column that is ours is the first: no text we read *evaluated* a mechanism under the constraint that a previously published output is immutable, with per-recipient and coalition audit roles. That is absence of evidence after a bounded search, not a priority claim, and it would be false as a statement about what these methods *can* do — three of them were adapted into this setting in §8 and beat the mechanism developed here. The attribution ledger, naming every component this work reuses and from whom, is in `RELATED_WORK_COMPARISON.md`.

C Study 1: locked transport, full tables

Contrast	Endpoint	Estimate	Seeds 0 / 1 / 2	Adjusted 95%	
<i>F1 (registered primary): coalition penalty vs local controls, $c = 2.930$ over 20 endpoints</i>					
C1-L1	residence loss	+0.00029	+0.0005 / +0.0004 / -0.0001	[-0.00118, +0.00176]	
C1-L1	A / sex	-0.00282	-0.0046 / -0.0055 / +0.0017	[-0.00585, +0.00020]	
C1-L1	AB / sex	-0.00593	-0.0046 / -0.0094 / -0.0038	[-0.00915, -0.00271]	▼
C1-L1	A / race	-0.01369	-0.0145 / -0.0129 / -0.0137	[-0.01737, -0.01002]	▼
C1-L1	AB / race	-0.00792	-0.0054 / -0.0109 / -0.0075	[-0.01105, -0.00478]	▼
C1-L2	residence loss	+0.00059	+0.0012 / +0.0009 / -0.0004	[-0.00073, +0.00191]	
C1-L2	A / sex	-0.00258	-0.0020 / -0.0051 / -0.0007	[-0.00568, +0.00052]	
C1-L2	AB / sex	-0.00472	-0.0058 / -0.0060 / -0.0024	[-0.00804, -0.00140]	▼
C1-L2	A / race	-0.00741	-0.0054 / -0.0074 / -0.0095	[-0.01088, -0.00394]	▼
C1-L2	AB / race	-0.00302	+0.0004 / -0.0039 / -0.0056	[-0.00524, -0.00081]	▼
<i>F3 (registered secondary): coalition channel vs frozen neural channel J, $c = 2.728$ over 10 endpoints</i>					
C1-J	residence loss	-0.00932	-0.0141 / -0.0020 / -0.0118	[-0.01500, -0.00363]	▼
C1-J	A / sex	+0.01301	+0.0201 / +0.0034 / +0.0155	[+0.00753, +0.01848]	▲
C1-J	AB / sex	+0.00697	+0.0088 / +0.0014 / +0.0107	[+0.00151, +0.01242]	▲
C1-J	A / race	+0.03767	+0.0428 / +0.0383 / +0.0320	[+0.02941, +0.04594]	▲
C1-J	AB / race	+0.03268	+0.0330 / +0.0318 / +0.0332	[+0.02504, +0.04032]	▲

Family	Sens. cells	Better	Worse	Unresolved	advantage fired	c
F1 primary (C1 vs L1, L2; Mode B)	16	14	0	2	4 / 4	2.930
F2 replication (J vs L025, L20; Mode B)	16	6	0	10	4 / 4	3.024
F3 secondary (C1 vs J; Mode B)	8	0	8	0	0 / 2	2.728
F4 frozen transfer (all four; Mode A)	32	21	2	9	6 / 8	3.206

Contrast	Weight	Point est.	Pointwise 95% up.	NI?	Simult. 95% up.	NI?
C1 - L1	unw.	+0.00029	+0.00112	no	+0.00166	no
C1 - L1	PWGTP	+0.00010	+0.00103	no	+0.00164	no
C1 - L2	unw.	+0.00059	+0.00135	no	+0.00182	no
C1 - L2	PWGTP	+0.00017	+0.00102	no	+0.00162	no

Interface	source-probe allowance		half-headroom residence	
	Mode A (frozen)	Mode B (fresh)	Mode A	Mode B
H	3 / 3	3 / 3	0 / 3	0 / 3
E	3 / 3	3 / 3	0 / 3	0 / 3
A0	3 / 3	3 / 3	0 / 3	1 / 3
J	3 / 3	3 / 3	0 / 3	0 / 3
S0	0 / 3	3 / 3	2 / 3	1 / 3
L1	0 / 3	3 / 3	0 / 3	1 / 3
L2	0 / 3	3 / 3	0 / 3	1 / 3
C1	0 / 3	3 / 3	0 / 3	1 / 3

Comparator	gain at $p=1$	p^*	A/race at p^*	reachable?
E	0.0263	1.072	—	no
A0	0.0289	0.976	0.0638	yes
L025	0.0175	1.615	—	no
L20	0.0181	1.555	—	no
J	0.0189	1.493	—	no
S0	0.0304	0.929	0.0688	yes
C1 (target)	0.0282	—	0.0453	—

Interface	Residence gain	A / race		AB / sex	
		absolute	additional	absolute	additional
H	0.0000	0.0165	+0.0000	0.0141	+0.0000
J	0.0189	0.0241	+0.0076	0.0159	+0.0017
E	0.0263	0.0794	+0.0629	0.0299	+0.0158
A0	0.0289	0.0818	+0.0653	0.0393	+0.0251
S0	0.0304	0.0905	+0.0740	0.0397	+0.0256
L1	0.0285	0.0754	+0.0590	0.0288	+0.0146
L2	0.0288	0.0692	+0.0527	0.0275	+0.0134
C1	0.0282	0.0617	+0.0453	0.0228	+0.0087

D Study 2: attribution and denominators

Factor	Held at	Sens. cells	Better	Worse	Unres.	Residence better/worse
Nonlinear penalty	$r = 16$	24	7	0	17	0 / 0 (of 6)
Nonlinear penalty	$r = 8$	24	5	1	18	0 / 0 (of 6)
Rank-8 compression	both penalties	48	15	0	33	0 / 0 (of 12)
Coalition vs local penalty	both ranks	64	29	2	33	0 / 0 (of 16)

Endpoint	Weight	Estimate	Adjusted 95%
residence loss	unw.	-0.00435	[-0.01238, +0.00368]
A / sex	unw.	+0.01065	[+0.00331, +0.01799] ▲
AB / sex	unw.	+0.00247	[-0.00396, +0.00890]
A / race	unw.	-0.00160	[-0.01160, +0.00840]
AB / race	unw.	-0.00170	[-0.00832, +0.00492]
residence loss	PWGTP	-0.00432	[-0.01349, +0.00485]
A / sex	PWGTP	+0.01210	[+0.00441, +0.01978] ▲
AB / sex	PWGTP	+0.00422	[-0.00273, +0.01117]
A / race	PWGTP	+0.00133	[-0.00934, +0.01200]
AB / race	PWGTP	+0.00052	[-0.00713, +0.00818]

Seed	q	U : pos / neg / ≈ 0	L1 pos	L2 pos	C1 pos	r_+
0	128	32 / 0 / 96	32	32	32	16
1	128	32 / 0 / 96	32	32	32	16
2	128	32 / 0 / 96	32	32	32	16

Family	Contrast	Endpoint	Weight	Estimate	Adjusted 95%
C1	nlr16-L2-lin16-L2	A / sex	unw.	-0.00597 ▼	[-0.01009, -0.00184]
C1	nlr16-L2-lin16-L2	A / race	unw.	-0.01706 ▼	[-0.02477, -0.00935]
C1	nlr16-L2-lin16-L2	AB / race	unw.	-0.00956 ▼	[-0.01902, -0.00009]
C1	nlr16-L2-lin16-L2	A / race	PWGTP	-0.01316 ▼	[-0.02176, -0.00456]
C1	nlr16-C1-lin16-C1	A / sex	unw.	-0.00639 ▼	[-0.01056, -0.00221]
C1	nlr16-C1-lin16-C1	A / race	unw.	-0.00728 ▼	[-0.01422, -0.00034]
C1	nlr16-C1-lin16-C1	AB / race	unw.	-0.01161 ▼	[-0.02116, -0.00206]
C6	nlr8-L1-lin8-L1	A / race	unw.	+0.00862 ▲	[+0.00168, +0.01556]
C6	nlr8-L2-lin8-L2	A / sex	unw.	-0.00850 ▼	[-0.01237, -0.00463]
C6	nlr8-L2-lin8-L2	A / sex	PWGTP	-0.00634 ▼	[-0.01054, -0.00214]
C6	nlr8-C1-lin8-C1	A / sex	unw.	-0.00653 ▼	[-0.01099, -0.00207]
C6	nlr8-C1-lin8-C1	AB / sex	unw.	-0.00548 ▼	[-0.00858, -0.00238]
C6	nlr8-C1-lin8-C1	AB / sex	PWGTP	-0.00518 ▼	[-0.00876, -0.00159]
C3	lin8-L1-lin16-L1	A / race	unw.	-0.01721 ▼	[-0.02744, -0.00698]
C3	lin8-L1-lin16-L1	AB / race	unw.	-0.01808 ▼	[-0.02775, -0.00842]
C3	lin8-L1-lin16-L1	A / race	PWGTP	-0.01336 ▼	[-0.02493, -0.00178]
C3	lin8-L1-lin16-L1	AB / race	PWGTP	-0.01232 ▼	[-0.02325, -0.00140]
C3	nlr8-L1-nlr16-L1	A / race	unw.	-0.00935 ▼	[-0.01742, -0.00129]
C3	nlr8-L1-nlr16-L1	AB / race	unw.	-0.01133 ▼	[-0.02115, -0.00151]
C3	lin8-L2-lin16-L2	A / race	unw.	-0.02384 ▼	[-0.03547, -0.01221]
C3	lin8-L2-lin16-L2	AB / race	unw.	-0.01359 ▼	[-0.02501, -0.00218]
C3	lin8-L2-lin16-L2	A / race	PWGTP	-0.01811 ▼	[-0.03112, -0.00509]
C3	nlr8-L2-nlr16-L2	A / sex	unw.	-0.00453 ▼	[-0.00903, -0.00002]
C3	lin8-C1-lin16-C1	A / race	unw.	-0.02169 ▼	[-0.03196, -0.01143]
C3	lin8-C1-lin16-C1	AB / race	unw.	-0.01490 ▼	[-0.02559, -0.00421]
C3	lin8-C1-lin16-C1	A / race	PWGTP	-0.01667 ▼	[-0.02864, -0.00470]
C3	nlr8-C1-nlr16-C1	AB / sex	unw.	-0.00636 ▼	[-0.01085, -0.00187]
C3	nlr8-C1-nlr16-C1	A / race	unw.	-0.01463 ▼	[-0.02500, -0.00425]
C4	lin16-C1-lin16-L1	AB / sex	unw.	-0.00515 ▼	[-0.00911, -0.00119]
C4	lin16-C1-lin16-L1	A / race	unw.	-0.01066 ▼	[-0.01642, -0.00489]
C4	lin16-C1-lin16-L1	AB / race	unw.	-0.01427 ▼	[-0.01980, -0.00874]
C4	lin16-C1-lin16-L1	AB / sex	PWGTP	-0.00603 ▼	[-0.01069, -0.00137]
C4	lin16-C1-lin16-L1	A / race	PWGTP	-0.01203 ▼	[-0.01849, -0.00558]
C4	lin16-C1-lin16-L1	AB / race	PWGTP	-0.01502 ▼	[-0.02149, -0.00856]
C4	lin8-C1-lin8-L1	A / race	unw.	-0.01514 ▼	[-0.02051, -0.00976]
C4	lin8-C1-lin8-L1	AB / race	unw.	-0.01108 ▼	[-0.01564, -0.00653]
C4	lin8-C1-lin8-L1	A / race	PWGTP	-0.01535 ▼	[-0.02161, -0.00908]
C4	lin8-C1-lin8-L1	AB / race	PWGTP	-0.00991 ▼	[-0.01535, -0.00446]
C4	nlr16-C1-nlr16-L1	A / sex	unw.	-0.00636 ▼	[-0.01037, -0.00235]
C4	nlr16-C1-nlr16-L1	A / race	unw.	-0.01870 ▼	[-0.02509, -0.01231]
C4	nlr16-C1-nlr16-L1	AB / race	unw.	-0.02253 ▼	[-0.03232, -0.01273]
C4	nlr16-C1-nlr16-L1	A / sex	PWGTP	-0.00728 ▼	[-0.01191, -0.00265]
C4	nlr16-C1-nlr16-L1	A / race	PWGTP	-0.01865 ▼	[-0.02570, -0.01161]
C4	nlr16-C1-nlr16-L1	AB / race	PWGTP	-0.01700 ▼	[-0.02824, -0.00576]
C4	nlr16-C1-nlr16-L2	AB / sex	unw.	+0.00399 ▲	[+0.00034, +0.00764]
C4	nlr8-C1-nlr8-L1	A / sex	unw.	-0.00544 ▼	[-0.00884, -0.00204]
C4	nlr8-C1-nlr8-L1	AB / sex	unw.	-0.00702 ▼	[-0.01050, -0.00353]
C4	nlr8-C1-nlr8-L1	A / race	unw.	-0.02397 ▼	[-0.03285, -0.01510]
C4	nlr8-C1-nlr8-L1	AB / race	unw.	-0.01871 ▼	[-0.02632, -0.01110]
C4	nlr8-C1-nlr8-L1	A / sex	PWGTP	-0.00595 ▼	[-0.00977, -0.00212]
C4	nlr8-C1-nlr8-L1	AB / sex	PWGTP	-0.00645 ▼	[-0.01046, -0.00243]
C4	nlr8-C1-nlr8-L1	A / race	PWGTP	-0.02156 ▼	[-0.03232, -0.01081]
C4	nlr8-C1-nlr8-L1	AB / race	PWGTP	-0.01774 ▼	[-0.02684, -0.00864]
C4	nlr8-C1-nlr8-L2	A / sex	unw.	+0.00229 ▲	[+0.00009, +0.00449]
C4	nlr8-C1-nlr8-L2	AB / sex	unw.	-0.00428 ▼	[-0.00726, -0.00130]
C4	nlr8-C1-nlr8-L2	A / race	unw.	-0.00539 ▼	[-0.01044, -0.00033]
C4	nlr8-C1-nlr8-L2	AB / race	unw.	-0.00868 ▼	[-0.01433, -0.00302]
C4	nlr8-C1-nlr8-L2	AB / sex	PWGTP	-0.00385 ▼	[-0.00734, -0.00037]
C4	nlr8-C1-nlr8-L2	AB / race	PWGTP	-0.01018 ▼	[-0.01717, -0.00319]

E Study 3: registered forecasts, erasure rank, conditions

Seed	directions deleted	realised width	race class 3 support	cross-cov. after	idempotency	mean pres.
0	10	6	1	4.6×10^{-16}	4.9×10^{-15}	7.7×10^{-15}
1	10	6	1	3.9×10^{-16}	6.7×10^{-15}	4.9×10^{-15}
2	10	7	0	7.2×10^{-16}	3.0×10^{-15}	6.0×10^{-15}

Linear erasure deletes exactly $\text{rank}(\Sigma_{XZ})$ directions. A centred joint one-hot over sex (2), race (9) and coverage (2) has rank $1 + 8 + 1 = 10$, leaving 6 of 16. **Seed 2 differs because one race category has no support in that seed's own fitting pool**, so the centred one-hot has rank 9 and the channel keeps 7 dimensions. Support counts refer to fitting pools, not to the population: a category missing from one pool is not absent from California.

Interface	Objective	r	Fitting labels	Inf. inputs	Role
H	none (no channel)	–	none	H_A	reference
E	adversarial, frozen	16	sex, race	X	context
A0	adversarial, frozen	16	sex, race	X	context
L025	adversarial, frozen	16	sex, race	X	F2 control
L20	adversarial, frozen	16	sex, race	X	F2 control
J	adversarial, frozen	16	sex, race	X	neural ref.
S0	U only ($\lambda = 0$)	16	none	T, H_A	context
M025 / M1	$U - \lambda P_{\text{marg}}$	16	sex, race	T, H_A	context
L1 / L2 / L025	$U - \lambda P_{\text{loc}}$	16	sex, race, cov.	T, H_A	F1/F4 control
C025 / C1	$U - \lambda(P_{\text{loc}} + P_{AB})$	16	+ coal. sex, race	T, H_A	candidate
lin16-*	$U - \lambda P$ (alias)	16	as above	T, H_A	alias, not refitted
nlr16-*	nonlinear refinement	16	as above	T, H_A	dev. only
lin8-*	$U - \lambda P$, compressed	8	as above	T, H_A	dev. only
nlr8-*	nonlinear ref., compressed	8	as above	T, H_A	dev. only

F Attack scope, and what a paired contrast is invariant to

Condition	fresh independent attacks		+ historical catch-up (reporting scope)	
	absolute	additional	absolute	additional
H	0.0014	+0.0000	0.0071	+0.0000
J	0.0027	+0.0013	0.0024	−0.0047
LEACE	0.0053	+0.0039	0.0078	+0.0008
riv16-C1	0.0159	+0.0145	0.0159	+0.0088
A0	0.0342	+0.0328	0.0342	+0.0271

A / sex, 2018, unweighted, seed means. The reporting scope gives H and the frozen historical interfaces their saved-observer catch-up attacks; the new arms have none, so their *absolute* recovery is unchanged between scopes and only the *subtracted* H baseline moves ($0.0014 \rightarrow 0.0071$). Every paired contrast between two arms is therefore identical on both scales, because the shared baseline cancels; only H -relative levels and ratios are scope-dependent. The absolute recovery of J falls when it is given more attack candidates – a validation-selected maximum is a floor on leakage, never a bound.

G Pending integration

A fourth study—a direct adversarial neural refinement with frozen H , refreshed attackers, local and coalition controls, two widths and four penalty strengths—was in progress when this revision was completed and had committed no protocol. Its cells are left empty and marked pending; the preconditions we will check before integrating any of them are printed beside each row, and no forecast is entered in the meantime. Full detail: PENDING_EXPERIMENT_INTEGRATION.md.

Reported quantity	Value	Interval	Precondition checked before integration
Adversarially refined arm – riv16-C1, four sensitive endpoints	<i>pending</i>	<i>pending</i>	no residence/commute label in representation selection
Adversarially refined arm – J , four sensitive endpoints	<i>pending</i>	<i>pending</i>	attacker view equals the recipient view; H_B not in the release to A
Coalition vs. local controls, coordination cells	<i>pending</i>	<i>pending</i>	candidate-selection multiplicity covers the whole search
Residence loss vs. J and vs. riv16-C1	<i>pending</i>	<i>pending</i>	training attackers distinct from fresh auditors
Penalty-strength and width ablations	<i>pending</i>	<i>pending</i>	optimiser repeats nested in three fixed anchors, not extra seeds
Erasure and single-attacker controls	<i>pending</i>	<i>pending</i>	executed cell count read from the committed matrix, not the plan

Cells are deliberately empty. They are filled only from a committed artifact whose hash and executed-unit count have been checked, and no forecast is entered in the meantime.

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